USER MANUAL & DEPLOYMENT GUIDE

PKM Urdu Assistant

An Urdu question-answering assistant for Punjab Police Khidmat Markaz services

پنجاب پولیس خدمت مرکز — اردو معلوماتی اسسٹنٹ

Independent project. Not an official Punjab Police product. Service information is taken from pkm.punjab.gov.pk and should be confirmed with a Khidmat Markaz or the helpline 04299211356.

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1 Introduction

The PKM Urdu Assistant answers everyday questions about Punjab Police Khidmat Markaz (PKM) services in Urdu: what a service costs, which documents to bring, how long it takes, and where the nearest centre is.

Punjab Police runs Khidmat Markaz (service centres) in all 36 districts of Punjab. Through a single window they issue character certificates, police verifications and driving licence services, and they register tenants, employees and police reports. The details people need before visiting, such as fees, documents and centre addresses, are published on the PKM website, but they are spread across many pages and tabs.

The assistant brings this information into one conversation. A citizen types a question in Urdu, or in Roman Urdu when Gemini is enabled, and gets a short, direct answer taken from the official website's content.

This manual is for two readers:

- **Users**: section 8 explains how to ask questions and what to expect.
- **Developers and administrators**: sections 5–7 and 9–11 cover installation, Gemini setup, deployment, updating the data and troubleshooting.

2 Purpose and Objectives

The problem

- **Scattered information.** Each of the 14 services has its own tab with documents, fees, timing and delivery, and centre details are split across 36 district tabs.
- **Wasted trips.** People often reach a centre without a required document, such as an attested copy or an affidavit, and must come back.
- **Language and typing habits.** Many people are more comfortable asking in Urdu, and many type Urdu in Roman script. Most existing tools expect English and exact keywords.
- **Busy helplines and counters.** Simple, repeated questions take staff time that could go to actual service.

Purpose

The purpose of the assistant is to **give citizens accurate, instant answers about Khidmat Markaz services in their own language**, so that they arrive prepared and spend less time finding information.

Objectives

Accuracy first

Answers come only from information published on the PKM website. The assistant says “I don't know” rather than guess.

Urdu by default

Questions and answers are in Urdu, with support for Roman Urdu and common spelling variations.

Answer what was asked

A question about the fee gets the fee, not a whole page of text. A question about Lahore gets Lahore's centres.

Free and easy to run

Runs on an ordinary laptop or on free cloud hosting, with no paid services required.

Easy to keep current

Scrapers and a cleaning script refresh the data when the website changes.

Measurable

A built-in evaluation set shows how accurate the assistant is after every change.

Intended users

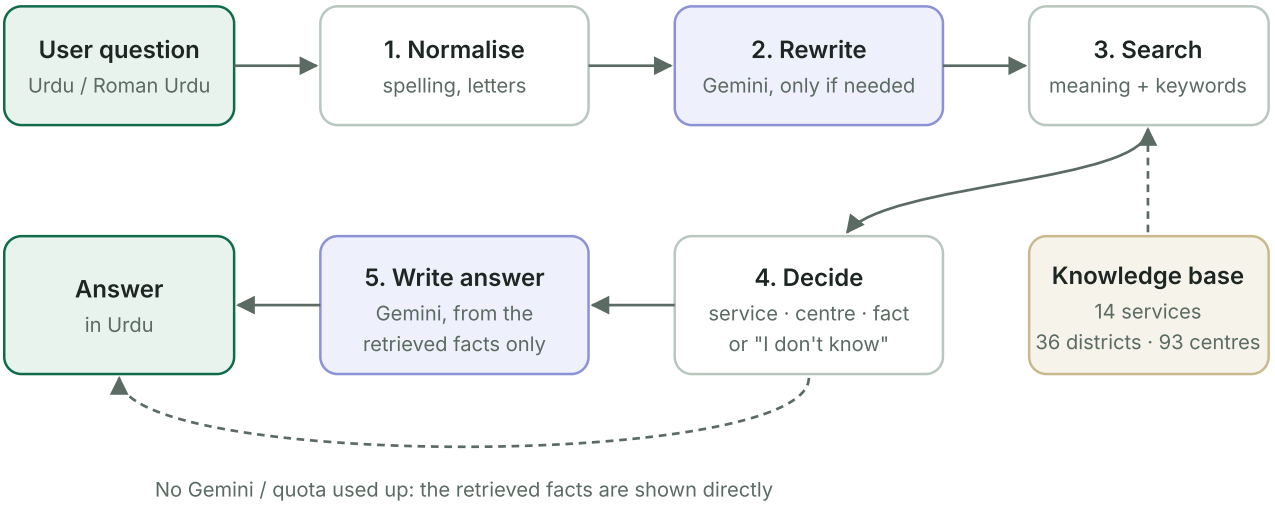
Citizens of Punjab who need a PKM service; family members applying on behalf of someone abroad; students and developers studying Urdu natural-language systems.

3 Key Features

Feature	What it means for the user
14 services	Character certificate, police verification, five driving licence services, tenant and employee registration, vehicle verification, lost item, crime and women-violence reports, and FIR copy.
36 districts, 93 centres	Address, phone numbers, email and opening hours for every Khidmat Markaz.
Focused answers	Asks about the fee? Gets the fee. Asks about documents? Gets the document list.
Understands spelling variations	سرٹیفکیٹ and سرٹیفیکیٹ, پنڈی and راولپنڈی, “renew” and تجدید are treated the same.
Natural answers (Gemini)	Short, conversational Urdu answers written from the official facts.
Roman Urdu (Gemini)	“character certificate ki fees kitni hai” works too.
Follow-up questions (Gemini)	After asking about a service, “اس کی فیس؟” (its fee?) is understood.
Honest refusals	Questions outside its knowledge get a polite refusal and a list of topics it covers.
Three ways to use it	Web app (phone or computer), desktop app, and terminal.
Feedback	👍 / 👎 buttons on each answer help improve the assistant.
Works offline	Without Gemini the assistant still answers in basic mode, showing the facts directly.

4 How the Assistant Works

The assistant uses an approach called **Retrieval-Augmented Generation (RAG)**. It first *retrieves* the relevant official facts, then (optionally) uses an AI model to *write* a natural answer from those facts only. The AI never answers from its own general knowledge.



Purple steps use Gemini and are optional. Without Gemini, steps 2 and 5 are skipped.

Step by step

1. **Normalise.** Spelling variants, Arabic-keyboard letters and diacritics are made uniform, so different ways of writing the same word match.
2. **Rewrite (Gemini, only when needed).** Roman Urdu, English or follow-up questions are rewritten into a complete Urdu question, e.g. “pindi ka khidmat markaz” becomes راولپنڈی خدمت مرکز کا پتہ.
3. **Search.** The question is converted into a numeric “meaning vector” by a multilingual language model and compared with 111 pieces of knowledge. In parallel, the assistant looks for district names, service names and words like فیس (fee), کاغذات (documents) or پتہ (address).
4. **Decide.** Using the search scores and keywords, the assistant chooses what to answer, as shown in the table below.
5. **Write the answer (Gemini).** Gemini receives the question, the chosen facts and the last few messages, with strict instructions to use only those facts. If Gemini is unavailable, the facts are shown directly.

How the assistant decides

Situation	Answer
A district is named, and the question is about address, phone or hours	That district's centres
Asks where a centre is, but no district named	Asks the user for their district
Best match is below the confidence threshold	Polite “I don't know” + list of topics (Gemini is not called)
General topic (complaints, tracking, verification)	The general information
A service plus a specific aspect (fee, documents, time, delivery, age)	Only that part of the service
A service without a specific aspect	The full service card

5 System Requirements

To run the assistant

Operating system	Windows 10/11, macOS 12+, or Linux
Python	3.10 – 3.14 (tested on 3.14)
Memory (RAM)	4 GB minimum, 8 GB recommended
Disk space	About 4 GB (PyTorch, the language model and packages)
Internet	Needed for installation and the first start; afterwards only for Gemini
Browser	Any modern browser for the web app (Chrome, Edge, Firefox, Safari)

Optional

Google account	For a free Gemini API key (section 7)
Hugging Face account	For free public hosting (section 9)
Google Chrome	Only for re-scraping the website (section 10)

Components used

Component	Role
paraphrase-multilingual-mpnet-base-v2	Free multilingual model that turns text into meaning vectors (runs locally)
Gemini (Flash) via google-genai	Optional: writes natural answers, understands Roman Urdu and follow-ups
Gradio 6	Web chat interface
Tkinter	Desktop interface (included with Python)
Selenium + BeautifulSoup	Website scraping

6 Installation

These steps are written for Windows. On macOS or Linux, use the commands marked "macOS/Linux".

1. **Install Python.** Download Python 3.11 from python.org. During installation, tick **"Add Python to PATH"**. Check it worked by opening Command Prompt and typing

```
python --version
```

2. **Extract the project.** Unzip the project to a simple path, for example `C:\Projects\Urdu_Chatbot`. Avoid paths with spaces or Urdu characters.

3. **Open a terminal in the folder.** In File Explorer, open the folder, click the address bar, type `cmd` and press Enter.

4. **Create a virtual environment** (keeps this project's packages separate):

```
python -m venv venv
venv\Scripts\activate          (Windows)
source venv/bin/activate       (macOS/Linux)
```

   You should now see `(venv)` at the start of the line. Activate it again every time you open a new terminal.

5. **Install the packages.** This takes 5–15 minutes the first time.

```
pip install -r requirements.txt
```

6. **Prepare the data.** The cleaned data is already included; run this only after re-scraping.

```
python clean_data.py
```

7. **Start the web app.**

```
python app.py
```

   Open **http://127.0.0.1:7860** in your browser. The first start downloads the language model (about 1 GB) and builds the search index, which can take a few minutes. Later starts take seconds.

> **Check that everything works**
>
> Run `python evaluate.py` . It asks 30 known questions and prints how many were answered correctly.

7 Setting Up Gemini (Optional)

The assistant works without Gemini, but Gemini makes it noticeably better: natural answers, Roman Urdu and follow-up questions. Gemini's free tier is enough for testing and small deployments.

1. **Get a free API key.** Go to **aistudio.google.com/apikey**, sign in with a Google account and click **Create API key**. Copy the key.
2. **Create the `.env` file.** In the project folder, copy `.env.example` and rename the copy to `.env`. Open it in Notepad and paste your key:

   ```
   GEMINI_API_KEY=paste-your-key-here
   ```

3. **Restart the app.** The status line at the bottom of the page should now read Gemini (gemini-flash-latest) (Gemini is active). فعال ہے

> **Keep your key secret**
>
> Never share the `.env` file, upload it to GitHub or send it to anyone. On Hugging Face, the key goes into an encrypted **Secret** instead (section 9).

> **Free tier and privacy**
>
> Free-tier request limits are shown in Google AI Studio. If they are used up, the assistant keeps working in basic mode and Gemini returns when the limit resets. On the free tier, Google may use prompts to improve its products, so the assistant should never be given personal information such as CNIC numbers.

Choosing a different model (optional)

By default the assistant uses `gemini-flash-latest`, Google's alias for its newest Flash model. To pin a specific model, add a line to `.env`: `GEMINI_MODEL=model-name`.

8 Using the Assistant

8.1 The web app



Figure 1. The web app when it opens (real screenshot). Numbers refer to the table below.

#	Part	What it does
1	Title	Name of the assistant.
2	Description	What you can ask, and that Urdu and Roman Urdu both work.
3	Conversation	Your questions (right) and the assistant's answers (left). Scrolls as the chat grows.
4	Question box	Type your question and press **Enter**.
5	Send (بھیجیں)	Same as pressing Enter.
6	Examples (مثال سوالات)	Ready-made questions; click one to put it in the question box.
7	New chat (نئی گفتگو)	Clears the conversation and starts fresh (also resets follow-up memory).
–	Answer buttons	Below each answer (Figure 2): copy, 👍 and 👎. Above the chat: copy all, clear and share.
–	Footer	Below the page: shows that this is unofficial, gives the helpline, and shows whether Gemini is active.

8.2 What you can ask

If you ask...	You get...
کریکٹر سرٹیفیکیٹ کی فیس کتنی ہے؟	Only the fee for Character Certificate.
ڈرائیونگ لائسنس کی تجدید کے لیے کون سے کاغذات چاہییں؟	The list of required documents for licence renewal.
کریکٹر سرٹیفیکیٹ	The full service card: documents, fee, processing time and delivery.
لاہور خدمت مرکز کا پتہ	All Lahore centres with address, phone, email and opening hours.
خدمت مرکز کہاں ہے؟	The assistant asks which district you are in.
پولیس کے خلاف شکایت کہاں کریں؟	General information: complaints can be filed at any Khidmat Markaz.
character certificate kitne din mein milta hai	Processing time (Roman Urdu needs Gemini).
اس کی فیس؟	A follow-up: the fee of the service you just asked about (needs Gemini).
پچھلا سوال کیا تھا؟	Repeats your previous question.
آج موسم کیسا ہے؟	A polite “I don't have this information” plus a list of topics it covers.

8.3 Tips for best results

- **Name the service.** کریکٹر سرٹیفیکیٹ کی فیس works better than just فیس کتنی ہے.
- **Name your district** when asking about a centre's address, phone or hours.
- **Ask one thing at a time** for the most focused answer. Gemini can also combine parts if you ask for “fee and documents” together.
- **Never type personal details** such as CNIC, passport or phone numbers. The assistant does not need them.
- **Confirm before you travel.** Fees and requirements can change; the helpline is 04299211356.

8.4 Example answers



Figure 2. A fee question returns only the fee, and a documents question only the list of required documents for licence renewal. Gemini wrote both answers from the retrieved facts (real screenshot).

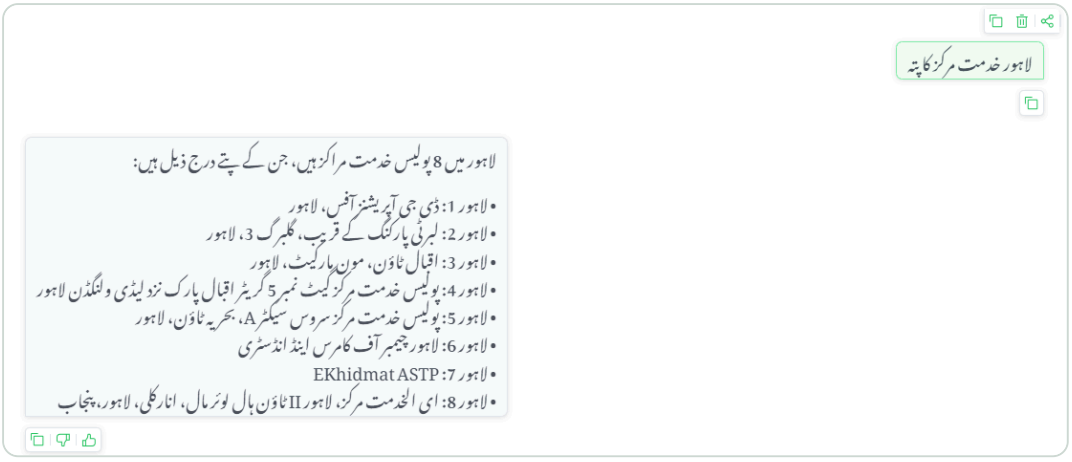


Figure 3. A district question lists all 8 Lahore centres with their addresses (real screenshot).

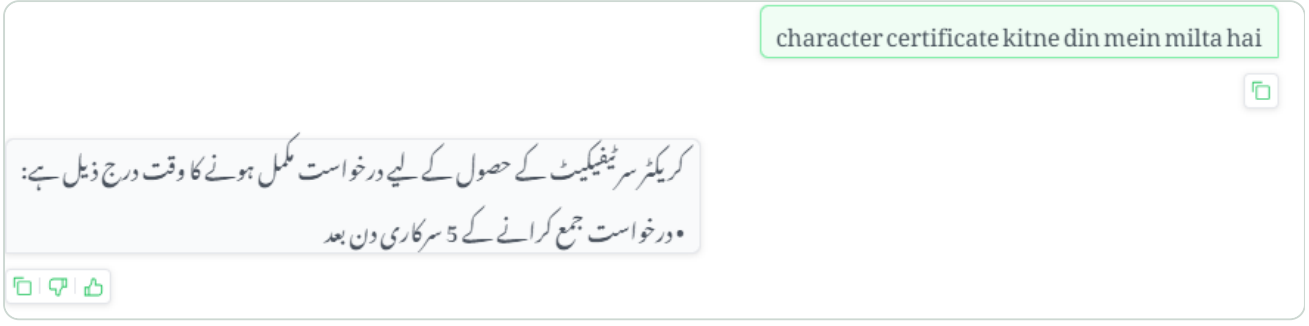


Figure 4. A question typed in Roman Urdu is understood and answered in Urdu script (real screenshot).

8.5 Desktop app and terminal

The same assistant can run without a browser:

```
python urdu_chatbot.py          desktop window (Tkinter)
python urdu_chatbot.py --cli    terminal: also shows the source and score of each answer
```

In the desktop app, type a question and press Enter. **ہسٹری** shows the last 20 questions and answers; **صاف کریں** clears the window. The terminal mode is the fastest way for developers to see *why* an answer was chosen.

8.6 Feedback and history

Every question and answer is saved to `data/chat_history.jsonl`, and every 👍/👎 to `data/feedback.jsonl`. Section 10 explains how to use them to improve the assistant.

9 Deployment

Choose a deployment option based on who needs to use the assistant and for how long.

Option	Best for	Cost	Link lifetime
A. Local	Development, your own use	Free	Only on your computer
B. Share link	A quick demo to a teacher or friends	Free	About 1 week, only while your PC is on
C. Hugging Face Spaces (recommended)	A permanent public link, portfolio	Free	Permanent
D. Own server (VPS)	Custom domain, full control	Paid	Permanent

9.1 Option B: Temporary share link

```
python app.py --share
```

Gradio prints a public link ending in `.gradio.live`. Anyone can open it while the app is running on your computer. Closing the terminal or shutting down the PC ends the link.

9.2 Option C: Hugging Face Spaces with the deploy script (easiest)

Hugging Face Spaces hosts the app for free on a CPU server (2 vCPU, 16 GB RAM), which is enough for this assistant.

1. **Create an account** at huggingface.co and confirm your email.
2. **Create an access token.** Click your profile picture → **Settings** → **Access Tokens** → **Create new token**. Choose type **Write**, name it e.g. “deploy”, and copy it (it starts with `hf_`).
3. **Add the token to `.env`**, below your Gemini key:

```
GEMINI_API_KEY=your-gemini-key
HF_TOKEN=hf_your-token
```

4. **Deploy** (the Hugging Face library is already installed with the other packages):

```
python deploy_hf.py
```

   The script creates a Space named `pkm-urdu-assistant`, saves your Gemini key as an encrypted secret and uploads the project. Your `.env` file, caches and logs are never uploaded. Options: `--name other-name`, `--private`.

5. **Wait for the build.** Open the link the script prints, e.g. `https://huggingface.co/spaces/<your-name>/pkm-urdu-assistant`. The first build takes 5–10 minutes (status **Building**, then **Running**). Follow progress in the **Logs** tab.

6. **Share the link.** Anyone can now use the assistant from a phone or computer.

9.3 Option C: Hugging Face Spaces through the website (no script)

1. Go to **huggingface.co/new-space**. Enter a Space name, choose SDK **Gradio**, hardware **CPU basic · Free**, visibility **Public**, and click **Create Space**.

2. Open the **Files** tab → **Add file** → **Upload files**. Upload all project files and folders (including `data/clean/`). **Do not upload** `.env`. Click **Commit changes**.

3. Open **Settings** → **Variables and secrets** → **New secret**. Name: `GEMINI_API_KEY`, value: your key. Save. The Space restarts automatically.

4. Wait for the build to finish, then open the **App** tab.

> **Why does the Space know what to run?**
>
> The block at the top of `README.md` (between the `---` lines) tells Hugging Face to use Gradio 6.31, Python 3.11 and to start `app.py`. Keep that block when editing the README.

9.4 Updating a deployed Space

After changing code or data, run `python deploy_hf.py` again (or upload the changed files through the website). The Space rebuilds automatically.

> **Things to know about free Spaces**
>
> - A Space with no visitors for 48 hours is **paused**. Open the Space page and click **Restart**.
> - Files written while running, such as chat history and feedback, are **lost on restart**. Download them regularly from the running Space, or keep logs locally.
> - Changing a secret restarts the app.

9.5 Option D: Your own server (advanced)

On any Linux VPS with 4 GB+ RAM, run steps 4–7 from section 6, then start the app on all network interfaces behind a web server such as Nginx (with HTTPS):

```
SPACE_ID=server python app.py --port 7860     # SPACE_ID makes it listen on 0.0.0.0
```

Use a process manager (systemd, pm2 or a Docker container) so the app restarts automatically after a reboot.

10 Maintaining and Updating

10.1 Refreshing the data from the website

Run this whenever PKM updates fees, documents or centres. Google Chrome must be installed; the right driver is downloaded automatically.

```
python extract_services.py      services  -> data/raw/extracted_services.json
python extract_one.py           centres   -> data/raw/extracted_data.json
python extract_whole.py         home page -> data/raw/extracted_data_pkm.json
python clean_data.py            raw -> data/clean/
```

Add `--show` to any scraper to watch the browser. `clean_data.py` prints a summary and warns about anything it could not read. The general facts (complaints, tracking, head office) are maintained by hand in the `general_info()` function of `clean_data.py`. Re-check them against the home page output.

Restart the app afterwards. It notices the data changed and rebuilds the search index automatically.

10.2 Measuring accuracy

```
python evaluate.py
```

The script asks every question in its test list, marks each PASS or FAIL and prints the total. It also prints two numbers that guide the **threshold**, the minimum similarity score needed to answer:

- **lowest in-scope score**: the weakest match among questions the assistant *should* answer;
- **highest out-of-scope score**: the strongest match among questions it should *refuse*.

Set the threshold between the two: either `DEFAULT_THRESHOLD` in `chatbot_core.py`, or `--threshold 0.5` when starting the app.

> **Current result**
>
> On the current test set of 33 questions the assistant answers **33/33** correctly (first run: 26/30). The full evaluation matrix is in the Project Report.

10.3 A simple improvement routine (weekly)

1. Open `data/feedback.jsonl` and look at answers marked 👎.
2. Search `data/chat_history.jsonl` for معاف کیجیے: questions the assistant could not answer.
3. For each real problem, add the question and its expected answer to `CASES` in `evaluate.py`.

4. Fix the cause: a new spelling in `SEARCH_VARIANTS`, a district alias in `DISTRICT_ALIASES`, a keyword in `ASPECT_WORDS`, or missing data.
5. Run `python evaluate.py`. The score should go up and nothing that passed before should fail.

11 Troubleshooting

Problem	Solution
`ModuleNotFoundError` when starting	Activate your virtual environment, then run `pip install -r requirements.txt`.
First start is very slow	Normal. The embedding model (about 1 GB) downloads once and the search index is built once. Later starts take seconds.
Status line says :بند Gemini GEMINI_API_KEY not set	No API key found. Create `.env` with `GEMINI_API_KEY=...` (locally) or add the Space secret (Hugging Face), then restart.
Answer ends with جواب اس وقت AI) دستیاب نہیں...)	Gemini could not be reached. The terminal shows the reason on a line starting with `[gemini]`: quota used up (429), invalid key (400/403), model not found (404) or no internet. The facts shown are still correct.
A Roman Urdu question gets براہ کرم اپنا سوال اردو میں لکھیں (Figure 5)	Roman Urdu needs Gemini, so this means Gemini is not working. Check the `[gemini]` lines in the terminal. Until it is fixed, type the question in Urdu script.
Terminal shows `504 DEADLINE_EXCEEDED`, `503` or `429`	Google's servers were busy or slow. The app retries once and then switches automatically to a faster model (gemini-flash-lite-latest, then gemini-2.5-flash); the terminal shows `[gemini] using …`. If it keeps happening, try again later or set `GEMINI_MODEL=gemini-flash-lite-latest` in `.env`.
Status says بند Gemini with “404” or “not found”	The model name or API version is not available for your key. Add `GEMINI_MODEL=gemini-2.5-flash` (or another model listed in AI Studio) to `.env`. The app uses API version v1beta; change it with `GEMINI_API_VERSION` only if Google says so.
Says “I don't know” to a valid question	Run `python evaluate.py`. If in-scope scores sit just below the threshold, lower it slightly (`--threshold 0.4`). Add new spellings to `SEARCH_VARIANTS` in `text_utils.py`.
Wrong district or service detected	Add the missing spelling or English name to `DISTRICT_ALIASES` or `SEARCH_VARIANTS` in `text_utils.py`, then add the question to `evaluate.py`.
Urdu shows as boxes in the desktop app	Install an Urdu font such as Noto Nastaliq Urdu or Jameel Noori Nastaleeq. The web app loads its font automatically.
`Port 7860 is already in use`	Another copy is running. Close it, or start with `python app.py --port 7861`.

Hugging Face build fails	Open the Space's **Logs** tab. Usually a missing file (check `data/clean/` was uploaded) or a typo in `requirements.txt`.
Space shows “Paused”	Free Spaces pause after 48 hours without visitors. Click **Restart** on the Space page; it is back in a few minutes.
`deploy_hf.py`: 401 / Unauthorized	Your token is read-only or wrong. Create a token with **Write** permission and put it in `.env` as `HF_TOKEN`.
Scraper fails or returns nothing	Update Chrome. If the website layout changed, run with `--show` to watch it and adjust the selectors in the scraper.
`clean_data.py` warns about a new section	The website added a section. Add its Urdu heading to `SECTION_KEYS` in `clean_data.py` and matching words to `ASPECT_WORDS` in `chatbot_core.py`.

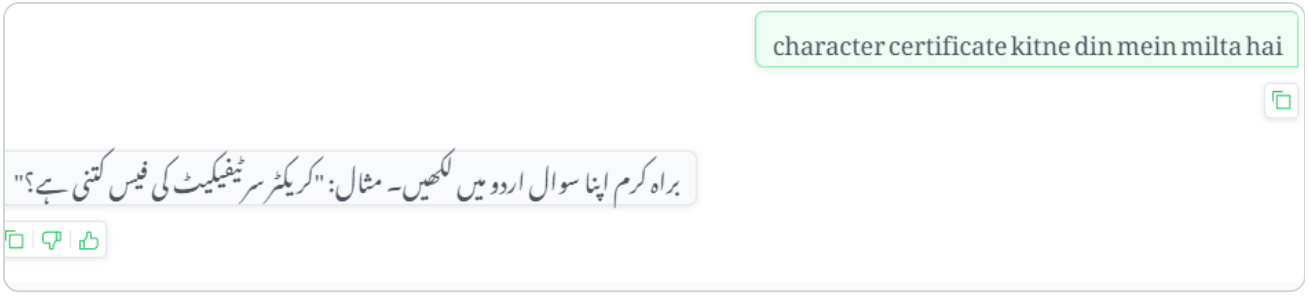


Figure 5. What a Roman Urdu question looks like when Gemini is not working (real screenshot). At start-up the app now makes one test request to Gemini, so the status line shows the real problem.

12 Frequently Asked Questions

Is this an official Punjab Police application?

No. It is an independent project that answers questions from information published on pkm.punjab.gov.pk. For final confirmation, contact a Khidmat Markaz or the helpline 04299211356.

Does it cost anything to run?

No. The embedding model runs on your own computer, Gemini has a free tier, and Hugging Face offers free CPU hosting.

Does it work without internet?

After the first setup, yes, in basic mode: the bot shows the retrieved facts directly. Gemini answers, Roman Urdu and follow-up questions need internet.

Can it submit an application or check my application status?

No. It only gives information. For status it explains how to use the official tracking page.

How up to date is the information?

It is as current as the last time the scrapers were run. See section 10 to refresh the data.

What happens to the questions people ask?

They are saved in data/chat_history.jsonl on the machine running the app, to help improve it. When Gemini is on, questions are also sent to Google, and on the free tier Google may use them to improve its products. Users should never type CNIC numbers or other personal details.

Can I add a new service or change an answer?

Yes. Re-scrape the website, or edit data/clean/services.json directly, then restart the app. The search index rebuilds automatically when the data changes.

13 Limitations and Disclaimer

Limitations

- **Information only.** The assistant cannot submit applications, take payments or check the status of a specific application.
- **Only as current as the last scrape.** If PKM changes a fee and the data is not refreshed, the assistant will give the old fee.
- **Limited scope.** It knows the 14 PKM services, the centres and a few general facts. Other police matters (traffic challans, emergencies, legal advice) are outside its knowledge, and it says so.
- **Basic mode is stricter.** Without Gemini, Roman Urdu and follow-up questions are not understood and answers are shown as plain fact lists.
- **AI wording.** Gemini is instructed to use only the retrieved facts, but its phrasing may occasionally be imperfect. The facts shown in basic mode are the reference.
- **Data gaps.** Some details are missing on the website itself (for example, the fee for driving licence verification), and opening dates for 11 districts could not be read reliably.

Disclaimer

The PKM Urdu Assistant is an independent project and is **not an official product of Punjab Police, PKM or the Punjab Information Technology Board**. Information is collected from the public website pkm.punjab.gov.pk and is provided as-is, without guarantee. Before visiting a centre or paying a fee, users should confirm the details with a Khidmat Markaz or the PKM helpline **04299211356**. In an emergency, call **15**.

Appendix A: Services Covered

The 14 services in the knowledge base, with the fee as listed on the PKM website at the time of the last scrape. Each service also has a list of required documents, the processing time and the delivery method; ask the assistant for details.

#	Service	Urdu name	Fee
1	Character Certificate	کریکٹر سرٹیفیکیٹ	500 روپے کسی بھی پی کے ایم کاؤنٹر پر جمع کرائی جا سکتی ہے۔
2	Police Verification	پولیس ویریفیکیشن	300 روپے کسی بھی پی کے ایم کاؤنٹر پر جمع کرائی جا سکتی ہے۔
3	Learner Driving Licence	لرنرز ڈرائیونگ لائسنس	کسی بھی پولیس خدمت مرکز سے 60 روپے کی ٹکٹ
4	Driving Licence Renewal	ڈرائیونگ لائسنس کی تجدید	موٹرسائیکل، رکشا: 500 روپے کار، ایل ٹی وی(پی ایس وی)، کمرشل ٹریکٹر: 750 روپے ایچ ٹی وی (پی ایس وی): 2000 روپے
5	International Driving Licence	بین الاقوامی ڈرائیونگ لائسنس	66 روپے عدالت کی ٹکٹ فی 450 روپے بنک چالان فی
6	Duplicate Driving Licence	نقل ڈرائیونگ لائسنس	موٹر سائیکل: 100 روپے کار، ایل ٹی وی: 300 روپے ایچ ٹی وی: 200 روپے
7	Driving Licence Verification	ڈرائیونگ لائسنس کی توثیق	Not listed on the website
8	Employee Registration	ملازم کی رجسٹریشن	مفت
9	Tenant Registration	کرائے دار کی رجسٹریشن	مفت
10	Vehicle Verification	گاڑی کی تصدیق	موٹر سائیکل: 500 روپے گاڑی، ایل ٹی وی، ایچ ٹی وی: 2000 روپے
11	Lost Item / Document Report	گمشدگی رپورٹ	مفت
12	Crime Report	جرائم رپورٹ	مفت
13	Violence Against Women Report	خواتین پر تشدد رپورٹ	مفت
14	FIR Copy	ایف آئی آر کی کاپی	مفت

Appendix B: Districts Covered

Khidmat Markaz information is available for all **36 districts** (93 centres). The number shows how many centres each district has. Opening hours on the website: پیر تا ہفتہ صبح 9 تا شام 5 بجے تک، اتوار بند.

بھکر 2	بہاولپور 3	بہاولنگر 2	اٹک 1
فیصل آباد 3	ڈی جی خان 3	چنیوٹ 1	چکوال 2
جھنگ 1	حافظ آباد 3	گجرات 2	گجرانوالہ 4
خوشاب 2	خانیوال 2	قصور 2	جہلم 3
منڈی بہاؤالدین 3	لودھران 1	لیہ 1	لاہور 8
ننکانہ صاحب 1	مظفر گڑھ 2	ملتان 1	میانوالی 4
رحیم یار خان 4	پاکپتن 1	اوکاڑہ 3	نارووال 2
سرگودھا 5	ساہیوال 2	راولپنڈی 8	راجن پور 2
وہاڑی 3	ٹوبہ ٹیک سنگھ 1	سیالکوٹ 4	شیخوپورہ 1

Appendix C: Command Reference

Command	What it does
`pip install -r requirements.txt`	Install all packages
`python app.py`	Start the web app at http://127.0.0.1:7860
`python app.py --share`	Web app + temporary public link
`python app.py --port 7861`	Use a different port
`python app.py --no-llm`	Run without Gemini (basic mode)
`python urdu_chatbot.py`	Desktop app
`python urdu_chatbot.py --cli`	Terminal mode with sources and scores
`--threshold 0.5`	Change the confidence threshold (app.py, urdu_chatbot.py, evaluate.py)
`python evaluate.py`	Accuracy test and threshold guidance
`python extract_services.py`	Scrape services
`python extract_one.py`	Scrape centres
`python extract_whole.py`	Scrape the home page
`python clean_data.py`	Clean raw data into data/clean/
`python deploy_hf.py`	Deploy or update on Hugging Face Spaces

Settings in .env

Variable	Purpose
`GEMINI_API_KEY`	Turns on Gemini answers
`GEMINI_MODEL`	Optional: a specific Gemini model (default gemini-flash-latest)
`HF_TOKEN`	Hugging Face write token, used only by deploy_hf.py

Appendix D: Project Files

File / folder	Purpose
`app.py`	Web app (Gradio); also what Hugging Face runs
`urdu_chatbot.py`	Desktop app and terminal mode
`chatbot_core.py`	The assistant's brain: search, decisions, answer formatting, history
`llm.py`	Gemini connection: question rewriting and answer writing
`text_utils.py`	Urdu spelling fixes, search variants, district aliases
`clean_data.py`	Turns raw scraped data into clean structured data
`extract_*.py`, `scraper_utils.py`	Website scrapers
`evaluate.py`	Test questions and accuracy report
`deploy_hf.py`	One-command Hugging Face deployment
`requirements.txt`	Python packages
`.env.example`	Template for your private settings (.env)
`README.md`	Quick start; its top block configures Hugging Face
`data/raw/`	Scraped data, as downloaded
`data/clean/`	services.json, centers.json, general_info.json (used by the assistant)
`data/cache/`	Saved search index (created automatically)
`data/*.jsonl`	Chat history and feedback logs (created automatically)